Operations on Singly Linked List

Q. Describe the fundamental operations performed on a Singly Linked List: Traversal, Insertion, Deletion, and Searching, detailing the logic behind each.

> **Definition to Remember**

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tempo

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1. Traversal & Searching

Traversal (Printing the list):

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Searching:

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2. Insertion Operations

| Position | Logic Steps |

| :--- | :

3. Deletion Operations

(Always check for Underflow: if `head == NULL`, list is empty).

| Position | Logic Steps |

| :--- | :

4. Critical Edge Cases

- **Empty List:** Trying to delete causes underflow.

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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Complete Executable C Code: Singly Linked List Full Implementation

```c
#inclu

struct Node {

int dat

// Insert at Head - O(1)

struct

// Insert at Tail - O(n)

struct

if (head == NULL) return newNode;

struct Node *temp = head;

while (

// Delete by Value - O(n)

struct

if (head->data == key) {

struct
```

```
struct Node *curr = head;
while (
if (curr->next != NULL) {
struct
// Traverse List
void tr
int main() {
struct
head = insertAtHead(head, 20);
head =
printf("\n--- Deleting Node 20 ---\n");
head =
return 0;
}
```